

AI Agentic Resume System for Students - Project Proposal

Problem Statement & Motivation

College students and recent graduates applying for internships and entry-level positions often submit dozens or even hundreds of applications. Each application ideally requires tailoring a resume and cover letter to match the specific job posting, but this process is time-consuming, repetitive, and mentally exhausting. As a result, students frequently submit generic resumes that fail to highlight relevant skills, reducing their chances of being selected by employers or applicant tracking systems (ATS).

The target users of this system are college students and early-career job seekers, particularly those applying to internships in competitive fields. An agentic AI solution is appropriate because resume tailoring involves multiple distinct but interdependent tasks: analyzing job descriptions, parsing resumes, matching skills, rewriting content, evaluating alignment, and tracking application history. These tasks are inefficient for humans to perform repeatedly at scale. A coordinated multi-agent system can autonomously manage this workflow, adapt outputs per job posting, and provide consistent feedback without requiring users to manually define or oversee each step.

Use Cases & Scenarios

Scenario 1: A student wants to tailor their resume for a specific internship posting. The system analyzes the job description, parses the student's master resume, identifies relevant skills and gaps, and generates a tailored resume version along with an explanation of the changes.

Scenario 2: A student wants to apply to multiple similar internship postings efficiently. The system groups similar job postings, creates optimized resume versions for each cluster, and produces application-ready materials while tracking which version was used for each employer.

Scenario 3: A student wants feedback on why they are not a strong match for certain roles. The system compares resume content to job requirements, highlights missing or weak skills, and suggests concrete improvements such as projects, coursework, or experience to strengthen future applications.

Initial System Design

The proposed system consists of several specialized agents working together:

- Job Analysis Agent: Extracts required skills, keywords, seniority level, and role expectations from job postings.
- Resume Parsing Agent: Structures and interprets resume content, identifying skills, experience, and metrics.
- Matching & Gap Analysis Agent: Compares resume content against job requirements and scores alignment.
- Rewrite Agent: Tailors resume bullet points and generates customized cover letters.
- ATS Optimization Agent: Ensures keyword coverage and flags formatting or phrasing risks.
- Critic Agent: Reviews generated content to prevent exaggeration or hallucinated experience.
- Application Tracker Agent: Stores resume versions and tracks which materials were used for each application.

Expected tools and data sources include large language model APIs, local file storage for resumes and application history, PDF and text parsing libraries, and a simple web-based front-end interface for user interaction.

Feasibility & Risks

Potential challenges include hallucination (nonexistent information) or overstatement of experience, which will be mitigated through a Critic Agent that enforces factual consistency.

Long job descriptions and resumes may introduce context-length constraints, requiring summarization or chunking strategies. Tool or API failures and rate limits may affect responsiveness but can be managed through caching and graceful error handling.

Privacy is a key constraint, as resumes contain sensitive personal data; therefore, the system will avoid unnecessary data retention and rely on local or secure storage.

Computational demands are moderate and suitable for a single-user academic project.

The project will be executed within the course duration, with milestones including proposal submission, single-agent prototype development, multi-agent workflow implementation, front-end integration, and final evaluation.